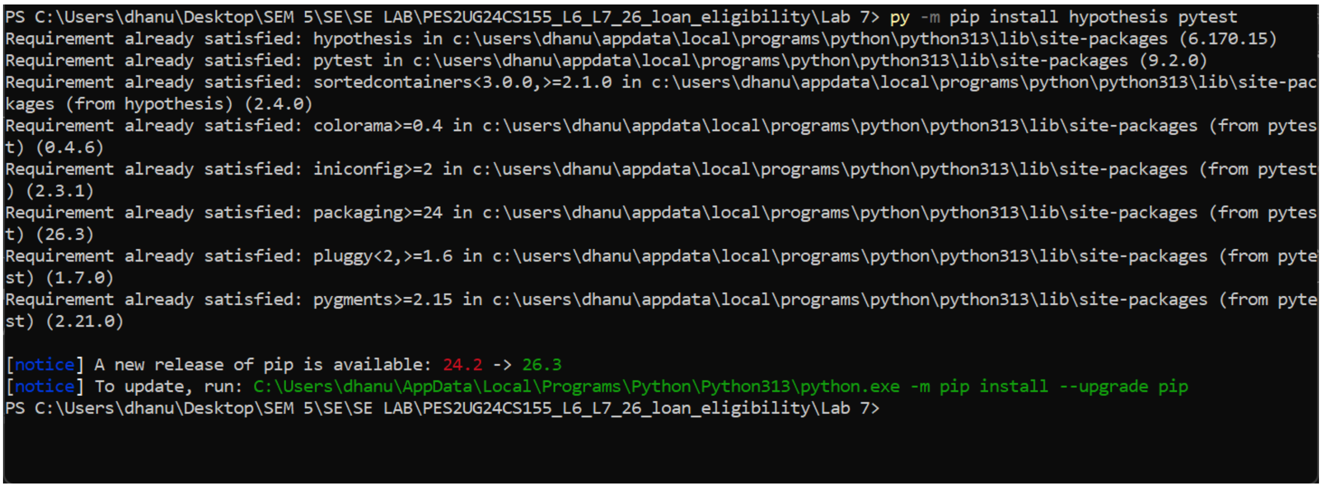
Lab 7: Fuzzing for Input Validation Bugs

SRN: PES2UG24CS155 | Repo: L6_L7_26_loan_eligibility (Lab 7)

Files: loan_eligibility.py (application), test_loan_eligibility.py (Hypothesis tests)

Step 1: Installing dependencies

Installed hypothesis and pytest using pip.

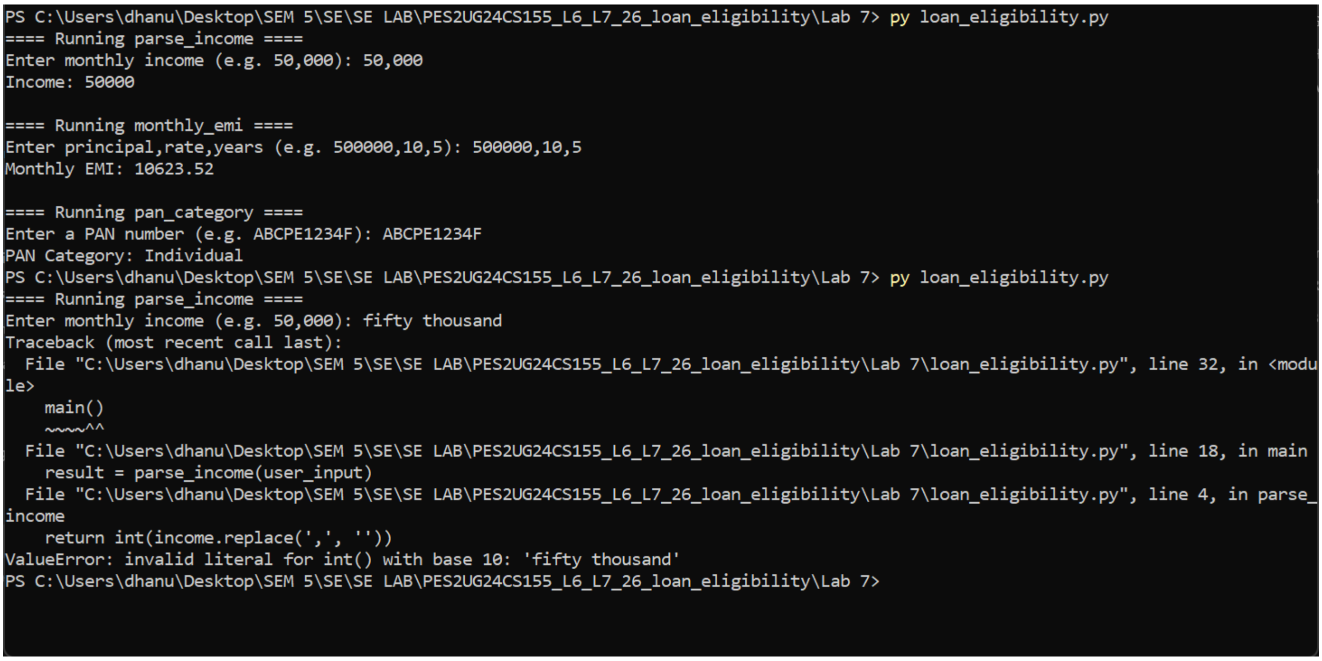


```
PS C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7> py -m pip install hypothesis pytest
Requirement already satisfied: hypothesis in c:\users\dhanu\appdata\local\programs\python\python313\lib\site-packages (6.170.15)
Requirement already satisfied: pytest in c:\users\dhanu\appdata\local\programs\python\python313\lib\site-packages (9.2.0)
Requirement already satisfied: sortedcontainers<3.0.0,>=2.1.0 in c:\users\dhanu\appdata\local\programs\python\python313\lib\site-pac
kages (from hypothesis) (2.4.0)
Requirement already satisfied: colorama>=0.4 in c:\users\dhanu\appdata\local\programs\python\python313\lib\site-packages (from pytes
t) (0.4.6)
Requirement already satisfied: iniconfig>=2 in c:\users\dhanu\appdata\local\programs\python\python313\lib\site-packages (from pytest
) (2.3.1)
Requirement already satisfied: packaging>=24 in c:\users\dhanu\appdata\local\programs\python\python313\lib\site-packages (from pytes
t) (26.3)
Requirement already satisfied: pluggy<2,>=1.6 in c:\users\dhanu\appdata\local\programs\python\python313\lib\site-packages (from pyte
st) (1.7.0)
Requirement already satisfied: pygments>=2.15 in c:\users\dhanu\appdata\local\programs\python\python313\lib\site-packages (from pyte
st) (2.21.0)

[notice] A new release of pip is available: 24.2 -> 26.3
[notice] To update, run: C:\Users\dhanu\AppData\Local\Programs\Python\Python313\python.exe -m pip install --upgrade pip
PS C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7>
```

Fig 1: pip install hypothesis pytest

Step 2: Running the buggy application

With the example inputs the program works. Typing a non-numeric income such as "fifty thousand" crashes it with a ValueError before it reaches the other two functions.



```
PS C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7> py loan_eligibility.py
==== Running parse_income ====
Enter monthly income (e.g. 50,000): 50,000
Income: 50000

==== Running monthly_emi ====
Enter principal,rate,years (e.g. 500000,10,5): 500000,10,5
Monthly EMI: 10623.52

==== Running pan_category ====
Enter a PAN number (e.g. ABCPE1234F): ABCPE1234F
PAN Category: Individual
PS C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7> py loan_eligibility.py
==== Running parse_income ====
Enter monthly income (e.g. 50,000): fifty thousand
Traceback (most recent call last):
  File "C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7\loan_eligibility.py", line 32, in <modu
le>
    main()
    ~~~~^^
  File "C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7\loan_eligibility.py", line 18, in main
    result = parse_income(user_input)
  File "C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7\loan_eligibility.py", line 4, in parse_
income
    return int(income.replace(',', ''))
ValueError: invalid literal for int() with base 10: 'fifty thousand'
PS C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7>
```

Fig 2: python loan_eligibility.py (original code)

Step 3: Writing the Hypothesis tests

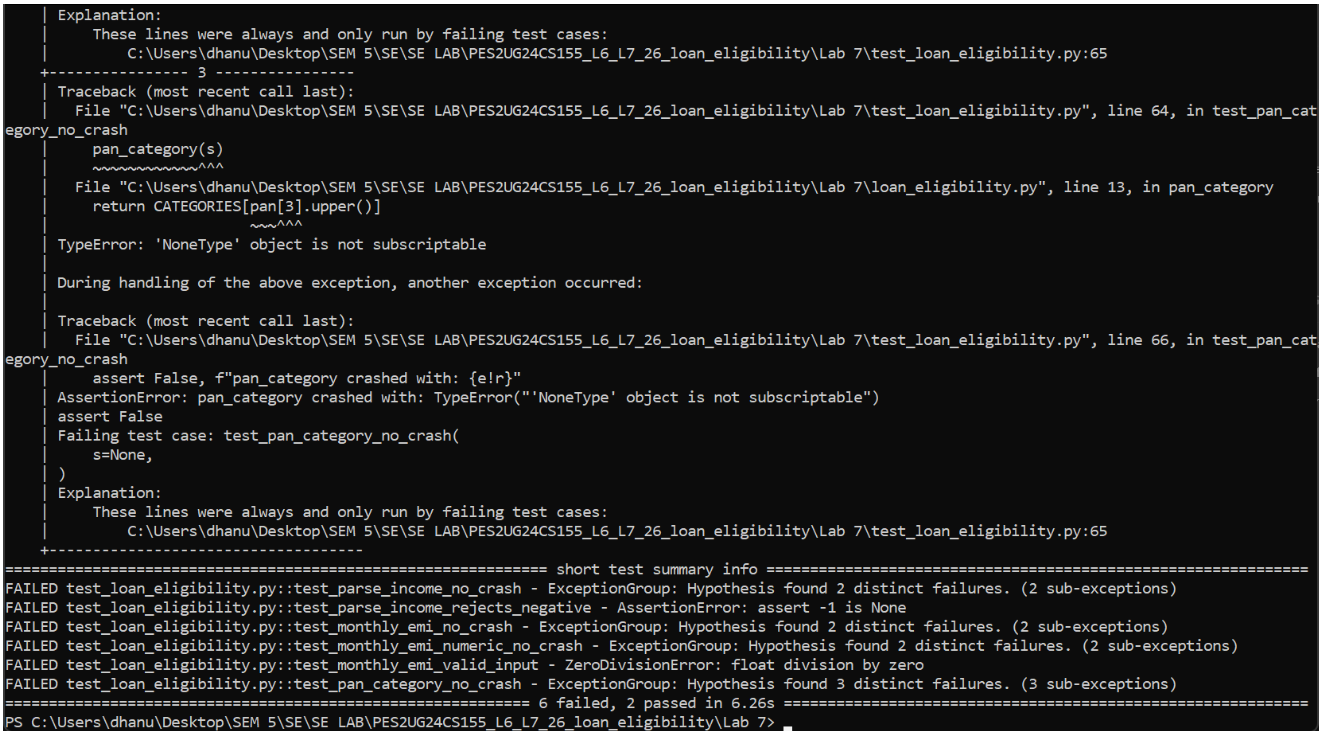
Each function has a no-crash test fed with st.text() **|** st.none(), as in the starter file. A few extra property tests were added to check that the functions also give correct results.

Test	Strategy	Property checked
test_parse_income_no_crash	st.text() | st.none()	never raises an exception
test_parse_income_formatted_number	st.integers(0, 10**12)	"50,000" style string gives back the same number
test_parse_income_rejects_negative	st.integers(max_value=-1)	negative income returns None
test_monthly_emi_no_crash	st.text() | st.none()	never raises an exception
test_monthly_emi_numeric_no_crash	3 x st.integers()	"p,r,y" with any integers (zero, negative, huge) never raises
test_monthly_emi_valid_input	realistic integer ranges	EMI is positive and EMI x months >= principal
test_pan_category_no_crash	st.text() | st.none()	never raises an exception
test_pan_category_valid_pan	st.from_regex(PAN format)	returns the category for the 4th character

Step 4: Running the tests on the buggy code

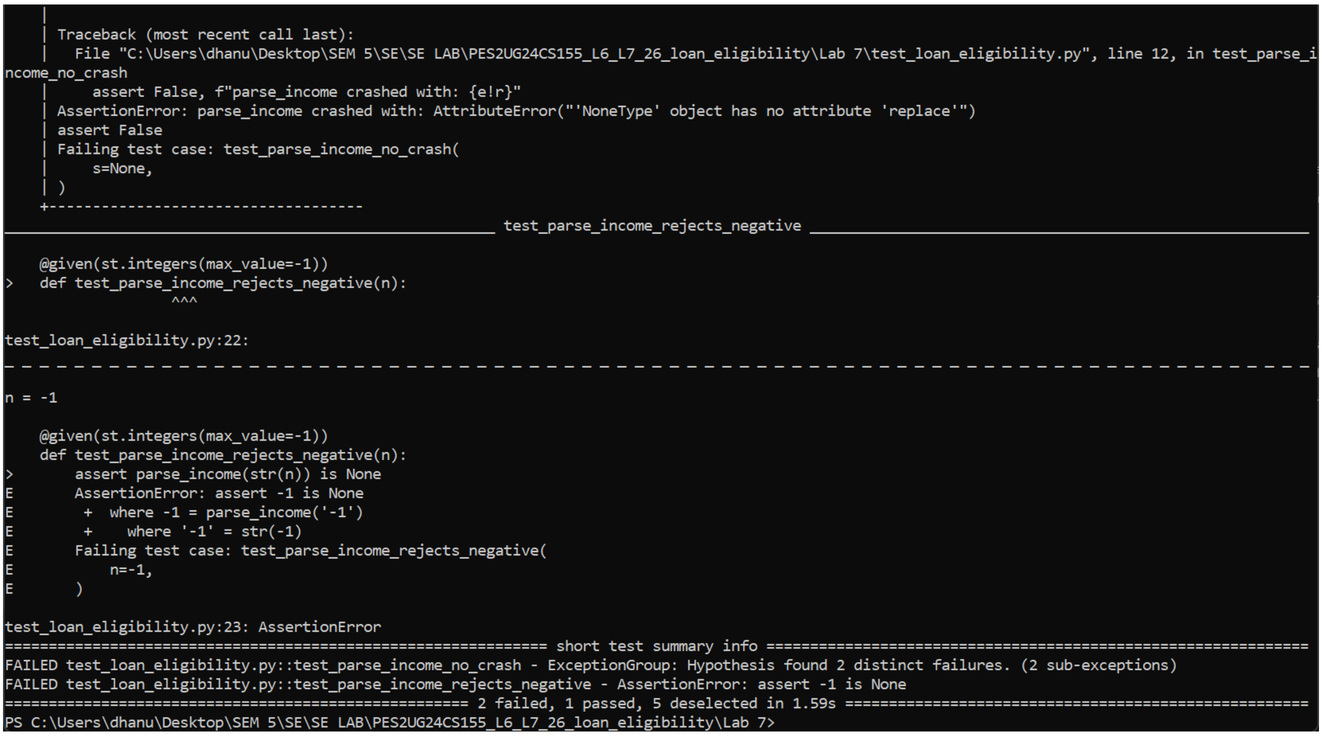
6 of the 8 tests fail. The minimal failing inputs Hypothesis found are listed below.

Function	Failing input	Result
parse_income	''	ValueError: invalid literal for int()
parse_income	None	AttributeError: no attribute 'replace'
parse_income	'-1'	returns -1 (negative income accepted)
monthly_emi	''	ValueError: not enough values to unpack
monthly_emi	None	AttributeError: no attribute 'split'
monthly_emi	'0,0,0' and '1000,0,1'	ZeroDivisionError (rate 0 or years 0)
monthly_emi	'0,-96,-710'	OverflowError: result too large
pan_category	''	IndexError: string index out of range
pan_category	'0000'	KeyError: '0'
pan_category	None	TypeError: 'NoneType' object is not subscriptable



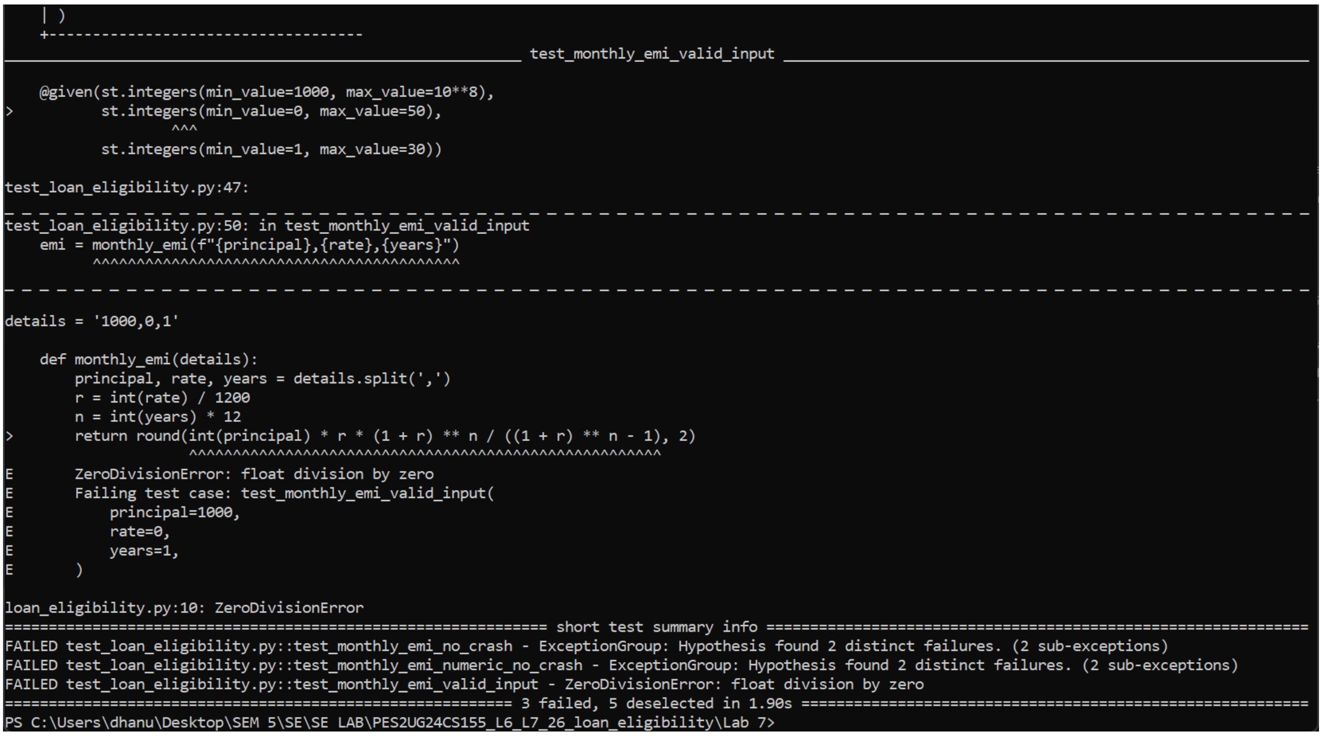
```
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7\test_loan_eligibility.py:65
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7\test_loan_eligibility.py", line 64, in test_pan_cat
egory_no_crash
    |     pan_category(s)
    |     ~~~~~~~~~~~~^^^
    |   File "C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7\loan_eligibility.py", line 13, in pan_category
    |     return CATEGORIES[pan[3].upper()]
    |                       ~~~^^^
    | TypeError: 'NoneType' object is not subscriptable
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7\test_loan_eligibility.py", line 66, in test_pan_cat
egory_no_crash
    |     assert False, f"pan_category crashed with: {e!r}"
    | AssertionError: pan_category crashed with: TypeError("'NoneType' object is not subscriptable")
    | assert False
    | Failing test case: test_pan_category_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7\test_loan_eligibility.py:65
    +------------------------------------
=============================================================== short test summary info ===============================================================
FAILED test_loan_eligibility.py::test_parse_income_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_loan_eligibility.py::test_parse_income_rejects_negative - AssertionError: assert -1 is None
FAILED test_loan_eligibility.py::test_monthly_emi_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_loan_eligibility.py::test_monthly_emi_numeric_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_loan_eligibility.py::test_monthly_emi_valid_input - ZeroDivisionError: float division by zero
FAILED test_loan_eligibility.py::test_pan_category_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================================================= 6 failed, 2 passed in 6.26s =============================================================
PS C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7>
```

Fig 3: pytest test_loan_eligibility.py, 6 failed, 2 passed



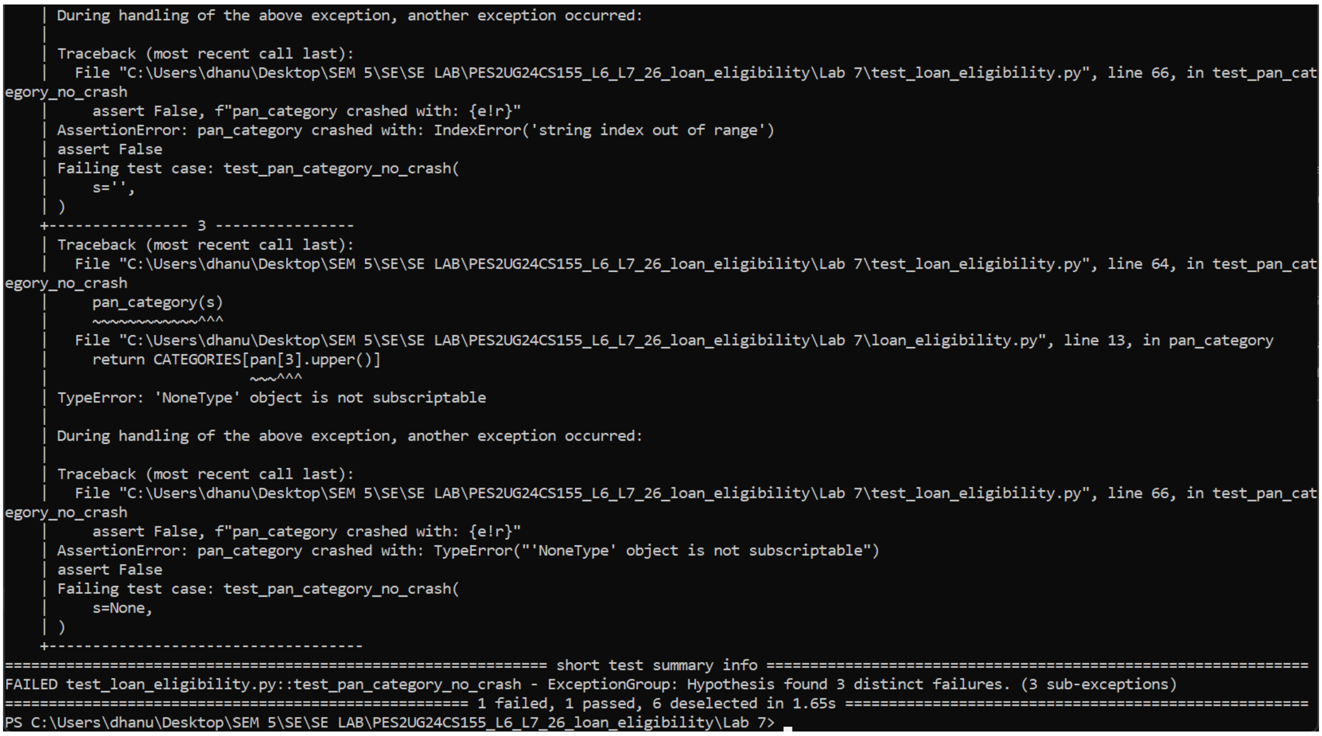
```
    |
    | Traceback (most recent call last):
    |   File "C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7\test_loan_eligibility.py", line 12, in test_parse_i
ncome_no_crash
    |     assert False, f"parse_income crashed with: {e!r}"
    | AssertionError: parse_income crashed with: AttributeError("'NoneType' object has no attribute 'replace'")
    | assert False
    | Failing test case: test_parse_income_no_crash(
    |     s=None,
    | )
    +------------------------------------
_________________________________________________________ test_parse_income_rejects_negative __________________________________________________________

    @given(st.integers(max_value=-1))
>   def test_parse_income_rejects_negative(n):
                   ^^^

test_loan_eligibility.py:22:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

n = -1

    @given(st.integers(max_value=-1))
    def test_parse_income_rejects_negative(n):
>       assert parse_income(str(n)) is None
E       AssertionError: assert -1 is None
E        +  where -1 = parse_income('-1')
E        +    where '-1' = str(-1)
E       Failing test case: test_parse_income_rejects_negative(
E           n=-1,
E       )

test_loan_eligibility.py:23: AssertionError
=============================================================== short test summary info ===============================================================
FAILED test_loan_eligibility.py::test_parse_income_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_loan_eligibility.py::test_parse_income_rejects_negative - AssertionError: assert -1 is None
====================================================== 2 failed, 1 passed, 5 deselected in 1.59s ======================================================
PS C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7>
```

Fig 4: failing parse_income tests (-k parse_income)



```
    | )
    +-----------------------------------
____________________________________________________________ test_monthly_emi_valid_input ____________________________________________________________

    @given(st.integers(min_value=1000, max_value=10**8),
>          st.integers(min_value=0, max_value=50),
                   ^^^
           st.integers(min_value=1, max_value=30))

test_loan_eligibility.py:47:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
test_loan_eligibility.py:50: in test_monthly_emi_valid_input
    emi = monthly_emi(f"{principal},{rate},{years}")
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

details = '1000,0,1'

    def monthly_emi(details):
        principal, rate, years = details.split(',')
        r = int(rate) / 1200
        n = int(years) * 12
>       return round(int(principal) * r * (1 + r) ** n / ((1 + r) ** n - 1), 2)
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       ZeroDivisionError: float division by zero
E       Failing test case: test_monthly_emi_valid_input(
E           principal=1000,
E           rate=0,
E           years=1,
E       )

loan_eligibility.py:10: ZeroDivisionError
============================================================= short test summary info ==============================================================
FAILED test_loan_eligibility.py::test_monthly_emi_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_loan_eligibility.py::test_monthly_emi_numeric_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_loan_eligibility.py::test_monthly_emi_valid_input - ZeroDivisionError: float division by zero
========================================================= 3 failed, 5 deselected in 1.90s ==========================================================
PS C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7>
```

Fig 5: failing monthly_emi tests (-k monthly_emi)



```
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7\test_loan_eligibility.py", line 66, in test_pan_cat
egory_no_crash
    |     assert False, f"pan_category crashed with: {e!r}"
    | AssertionError: pan_category crashed with: IndexError('string index out of range')
    | assert False
    | Failing test case: test_pan_category_no_crash(
    |     s='',
    | )
    +--------------- 3 ---------------
    | Traceback (most recent call last):
    |   File "C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7\test_loan_eligibility.py", line 64, in test_pan_cat
egory_no_crash
    |     pan_category(s)
    |     ~~~~~~~~~~~~^^^
    |   File "C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7\loan_eligibility.py", line 13, in pan_category
    |     return CATEGORIES[pan[3].upper()]
    |                       ~~~^^^
    | TypeError: 'NoneType' object is not subscriptable
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7\test_loan_eligibility.py", line 66, in test_pan_cat
egory_no_crash
    |     assert False, f"pan_category crashed with: {e!r}"
    | AssertionError: pan_category crashed with: TypeError("'NoneType' object is not subscriptable")
    | assert False
    | Failing test case: test_pan_category_no_crash(
    |     s=None,
    | )
    +-----------------------------------
============================================================= short test summary info ==============================================================
FAILED test_loan_eligibility.py::test_pan_category_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
=================================================== 1 failed, 1 passed, 6 deselected in 1.65s ====================================================
PS C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7>
```

Fig 6: failing pan_category tests (-k pan_category)

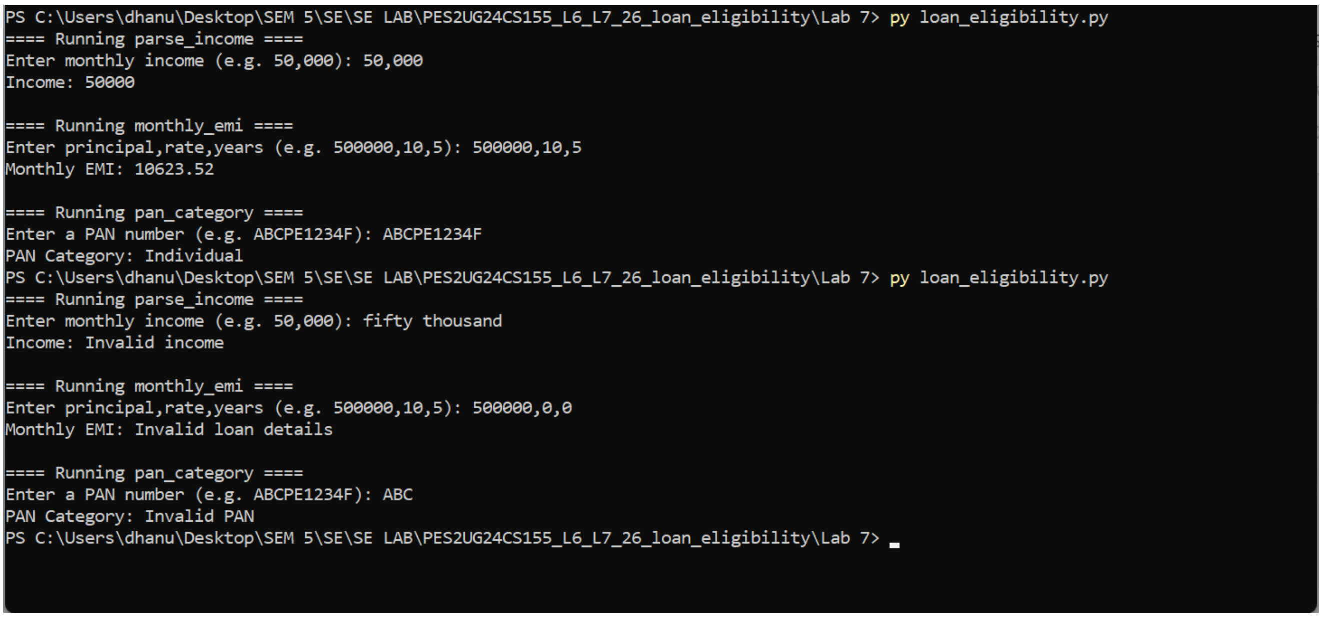
Step 5: Fixing the application

All three functions now return None for bad input instead of raising, and main() prints an "Invalid ..." message in that case.

parse_income: returns None if the input is not a string. After removing commas and spaces the value must be 1 to 15 ASCII digits (regex [0-9]+), so empty strings, words, decimals and negative numbers are rejected.

monthly_emi: input must be a string with exactly three comma separated parts, each a whole number. Principal must be above 0, rate between 0 and 100, years between 1 and 50. The year limit stops the overflow, and a 0% rate is handled separately as principal / months so there is no division by zero.

pan_category: input is stripped and upper-cased and must match the PAN format [A-Z]{5}[0-9]{4}[A-Z]. The category is looked up with CATEGORIES.get(), so an unknown 4th letter returns None instead of a KeyError.

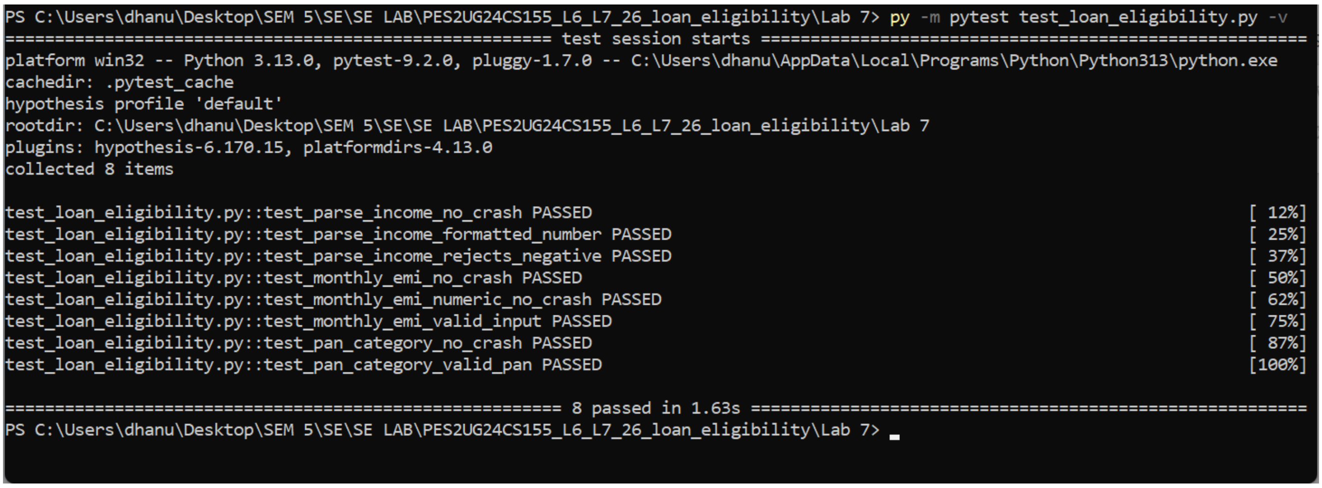


```
PS C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7> py loan_eligibility.py
==== Running parse_income ====
Enter monthly income (e.g. 50,000): 50,000
Income: 50000

==== Running monthly_emi ====
Enter principal,rate,years (e.g. 500000,10,5): 500000,10,5
Monthly EMI: 10623.52

==== Running pan_category ====
Enter a PAN number (e.g. ABCPE1234F): ABCPE1234F
PAN Category: Individual
PS C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7> py loan_eligibility.py
==== Running parse_income ====
Enter monthly income (e.g. 50,000): fifty thousand
Income: Invalid income

==== Running monthly_emi ====
Enter principal,rate,years (e.g. 500000,10,5): 500000,0,0
Monthly EMI: Invalid loan details

==== Running pan_category ====
Enter a PAN number (e.g. ABCPE1234F): ABC
PAN Category: Invalid PAN
PS C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7>
```

Fig 7: python loan_eligibility.py (fixed code), valid inputs then invalid inputs

Step 6: Re-running the tests on the fixed code

All 8 tests pass.



```
PS C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7> py -m pytest test_loan_eligibility.py -v
======================================================== test session starts =========================================================
platform win32 -- Python 3.13.0, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\dhanu\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7
plugins: hypothesis-6.170.15, platformdirs-4.13.0
collected 8 items

test_loan_eligibility.py::test_parse_income_no_crash PASSED                                                                      [ 12%]
test_loan_eligibility.py::test_parse_income_formatted_number PASSED                                                              [ 25%]
test_loan_eligibility.py::test_parse_income_rejects_negative PASSED                                                              [ 37%]
test_loan_eligibility.py::test_monthly_emi_no_crash PASSED                                                                       [ 50%]
test_loan_eligibility.py::test_monthly_emi_numeric_no_crash PASSED                                                               [ 62%]
test_loan_eligibility.py::test_monthly_emi_valid_input PASSED                                                                    [ 75%]
test_loan_eligibility.py::test_pan_category_no_crash PASSED                                                                      [ 87%]
test_loan_eligibility.py::test_pan_category_valid_pan PASSED                                                                     [100%]

========================================================= 8 passed in 1.63s ==========================================================
PS C:\Users\dhanu\Desktop\SEM 5\SE\SE LAB\PES2UG24CS155_L6_L7_26_loan_eligibility\Lab 7>
```

Fig 8: pytest test_loan_eligibility.py -v, 8 passed